



University
of Basel

Chapter 2: Applications

Foundations of Distributed Systems

Fall Semester 2025

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Prof. Dr. Isabel Wagner



Overview of Chapter 2

2	What are applications of parallel and distributed systems?
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2.1	... from a Communication Perspective
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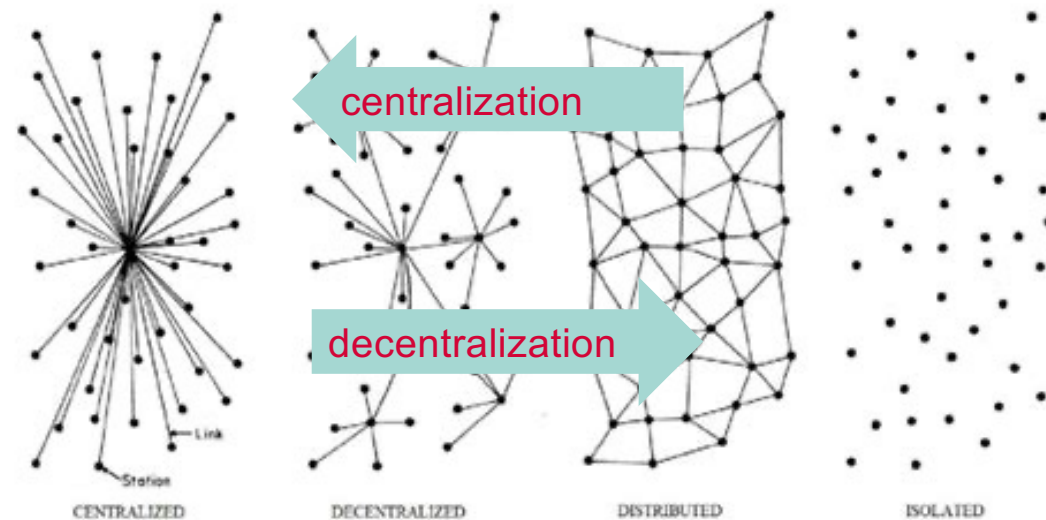
2.2	... from a Processing Perspective
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2.3	... from a Data Perspective
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2.4	... from a Privacy Perspective
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2.5	Mini-Seminar Topics
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(Physical and/or Logical) Structure of Distributed Systems



History of computing:
cycles alternating between centralization
and decentralization, influenced by
convenience, risk and economic factors. -
-> “**styles** of system distribution”

*Isn't every distributed system an “application” of
COMM? We pick two aspects:*

- a) networking (needed in both “styles”)*
- b) COMM as a decentralization tool box*

Image source: *On distributed communication: Intro to distr. Comm. networks* <https://apps.dtic.mil/sti/pdfs/AD0444830.pdf>

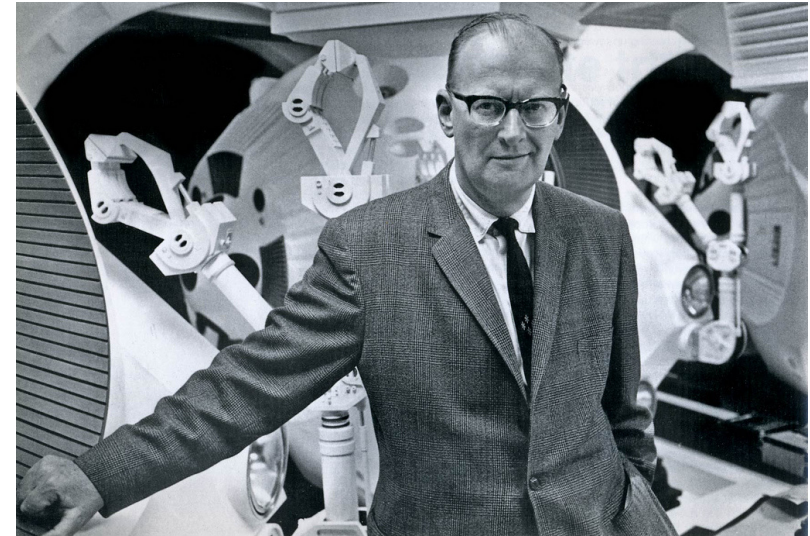
a) Networking – the Internet was predictable

Year 1964:

- *packet switching* concept just published in scientific literature
- 1964 is 10 years before a working TCP demo

We let the past speak about the future.

3 minutes video clip, featuring



“Sir **Arthur** Charles **Clarke** (1917–2008) was a British **science fiction writer**, science writer, futurist, inventor, undersea explorer, and television series host. He co-wrote the screenplay for the 1968 film *2001: A Space Odyssey*, widely regarded as one of the most influential films of all time.” (Wikipedia)

<https://youtu.be/1w8w8b1gEuI>

From “The Knowledge Explosion”, originally broadcast on BBC Two, Monday 21 September 1964

a) Networking – the Internet was predictable

Year 1964:

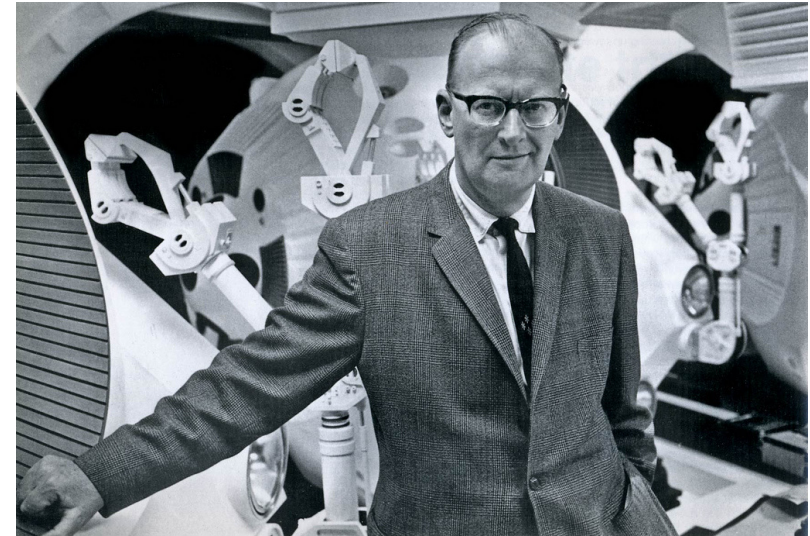
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In the news, just a few months ago:

Exclusive look at groundbreaking remote robotic surgery: Patient was in Africa; doctor was in Florida

The doctor calls it a "huge leap for health care."

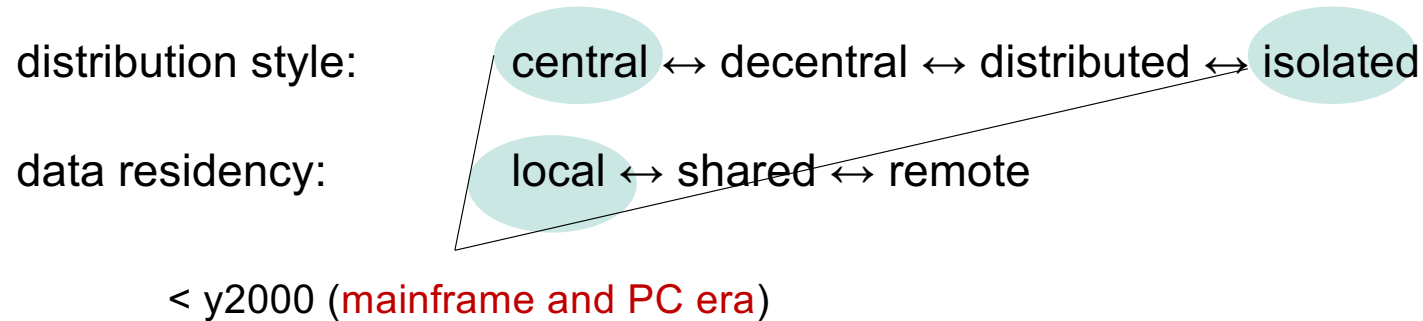
By [Youri Benadjaoud](#), [Osej Serratos](#), [Michael Mendelsohn](#), and [Milan Miller](#)

June 18, 2025, 1:11 AM



Dr. Vipul Patel, medical director of the Global Robotic Institute at Advent Health in Orlando, Florida, who performed the robotic telesurgery.

b) COMM concepts as a tool box



Historic context of around y2000, two developments:

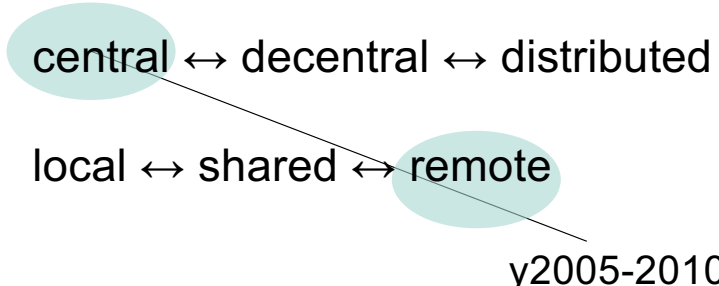
- research in **CSCW** (Computer-Supported Collaborative Work)
like multi-media needed several waves to become feasible and established
- novel “software as a service” concept (**SaaS**), in the pre-cloud era
SW does **not** run anymore on the PC, just the graphical user interface (=Web browser)

b) COMM concepts as a tool box (2/5)

distribution style: central ↔ decentral ↔ distributed

data residency: local ↔ shared ↔ remote

y2005-2010



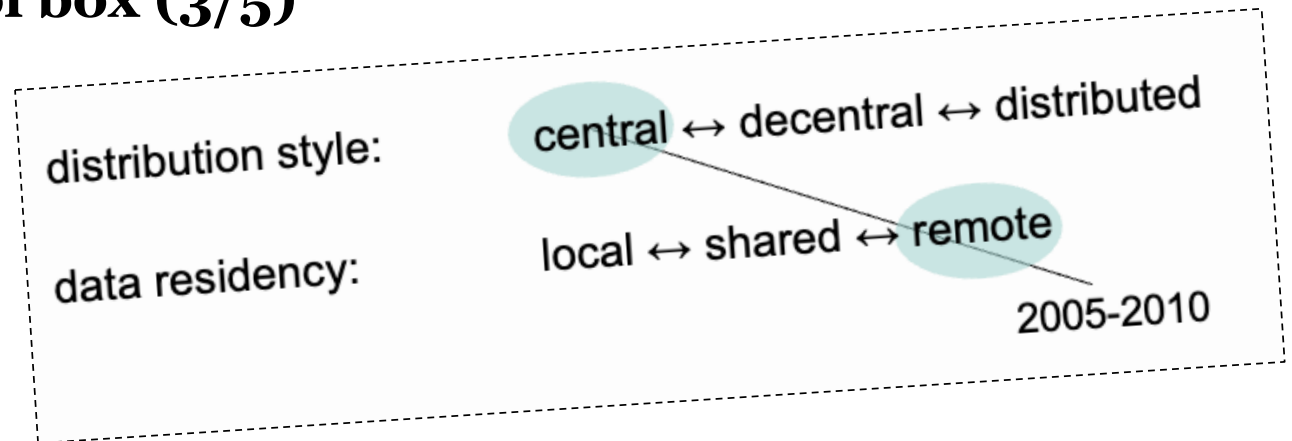
Example: Google Docs

- CSCW + SaaS: shared real-time editing
- major innovation: “Operational Transform” (OT)

keystrokes from a user are “transformed” before being sent to the other users
(case: insert a new character at a position which the other user just changed)



b) COMM concepts as a tool box (3/5)



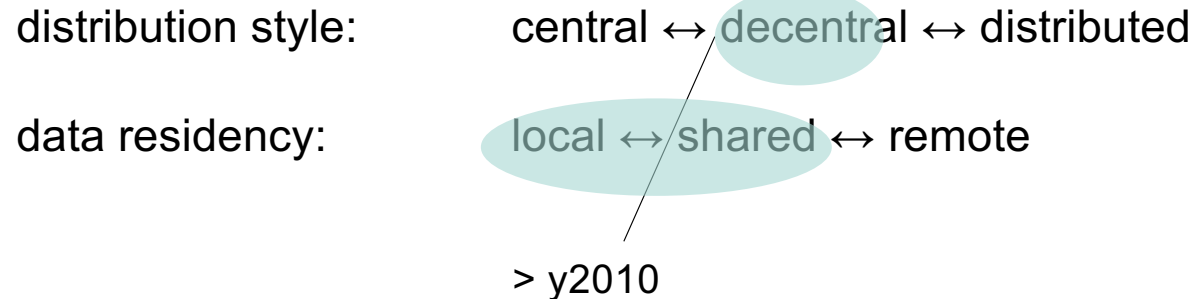
Google Docs - discussion

- highly dependent on Google's servers being online and available
- must trust Google to do the right thing (incl privacy concerns ...)

Are there alternatives?



b) COMM concepts as a tool box (4/5)



2011: Shapiro et al: Conflict-Free Replicated Data Types.

In: Défago et al (eds) *Stabilization, Safety, and Security of Distributed Systems*.

2011. Lecture Notes in Computer Science, vol 6976. Springer, Berlin, Heidelberg.

https://doi.org/10.1007/978-3-642-24550-3_29

- NOT dependent on single servers
- CRDTs much easier to implement than OT, but require some logical design effort
- many libraries available nowadays: Automerge, Yjs, ...

b) COMM concepts as a tool box (5/5)

CRDTs have been picked up by the big names
(Apple, Microsoft, Facebook etc)

Vision behind CRDTs:

programming (toolbox) view

**“[We want to] interact with state stored anywhere on a network as if it is a local variable, and program as if it is already downloaded and always up-to-date.”
(Michael Toomim, Braid protocol)**

structural view

**“shared global memory”
that also works when being offline
i.e., the network disappears**

Industry use [\[edit\]](#)

- [Zed](#) is an open-source collaborative IDE built by zed-industries that uses CRDTs for collaborative edits on the same file.
- [Fluid Framework](#) is an open-source collaborative platform built by [Microsoft](#) that provides server and client-side SDKs for creating modern real-time web applications using CRDTs.
- [Nimbus Note](#) is a collaborative note-taking application that uses the Yjs CRDT for collaboration.
- [Redis](#) is a distributed, highly available, and scalable in-memory database with a "CRDT" module for distributed data structures.
- [SoundCloud](#) open-sourced [Roshi](#), a LWW-element-set CRDT for the SoundCloud API.
- [Riak](#) is a distributed NoSQL key-value data store based on CRDTs.^[29] [League of Legends](#) uses Riak for its in-game chat system, which handles 7.5 million concurrent users and 11,000 rps.
- [Bet365](#) stores hundreds of megabytes of data in the Riak implementation of OR-Set.
- [TomTom](#) employs CRDTs to synchronize navigation data between the devices of a user.
- [Phoenix](#), a web framework written in [Elixir](#), uses CRDTs to support real-time multi-user editing.
- [Facebook](#) implements CRDTs in their Apollo low-latency "consistency at scale" data replication system.
- [Facebook](#) uses CRDTs in their FlightTracker system for managing the Facebook graph.
- Teletype for [Atom](#) employs CRDTs to enable developers share their workspace with others in real time.^[36]
- Haja Networks' OrbitDB uses operation-based CRDTs in its core data structure, IPFS.
- [Apple](#) implements CRDTs in the Notes app for syncing offline edits between multiple devices.
- [Swim](#) is a platform for running distributed real-time [streaming](#) applications that deliver updates to actors that stream pure op-based CRDT state updates to other actors in a DAG that represents the causal order of updates.
- [RxDB](#) is a client-side NoSQL database for distributed real-time [streaming](#) applications. It supports updating a document by storing NoSQL based CRDT deltas and replicating these with other nodes.
- [PGD](#) is a multi-master replication solution based on PostgreSQL. It supports [CRDT](#).

Summary “COMM applications”

- The Internet is here ... and will not go away. Is it in good shape?
- Unhealthy aspects:

- economic distortions towards centralization
- political threats: the real danger of a “Splinternet”



(decouple countries or continents from the global net)

→ *The Internet Archive has started to build and run data centers in Canada, Netherland, Switzerland*

- New research, products for “offline-first” mode, server-less operations

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Applications from a **Processing** Perspective ...

- From a processing perspective, ***we live in an HPC world***, where **supercomputing** is embraced by an increasing spectrum of classical and new fields and markets:



From Desktops, Laptops, Tablets, Phones, Watches, VR sets ...

iMac Pro

3.0 GHz 10-core (up to 18-core) Intel Xeon W processor



iPad Pro 4th gen

A12Z Bionic chip:
64-bit ARMv8
2.49 GHz
8-cores



iPhone 11 Pro

A13 Bionic chip (SoC):
64-bit ARMv8 2.66 GHz
6-cores CPU
4-cores GPU
8-cores Neural Engine



Watch Series 6

S6 SiP chip:
64-bit, 2-cores

Meta Quest 2

Qualcomm Snapdragon XR2 processor:
8-cores Kryo 585 CPU
(1×2.84, 3×2.43, 4×1.18 GHz),
Adreno 650 GPU
6GB RAM, 64-128 GB storage



MacBook Pro 16"

2.3 GHz 8-core Intel Core i9 processor

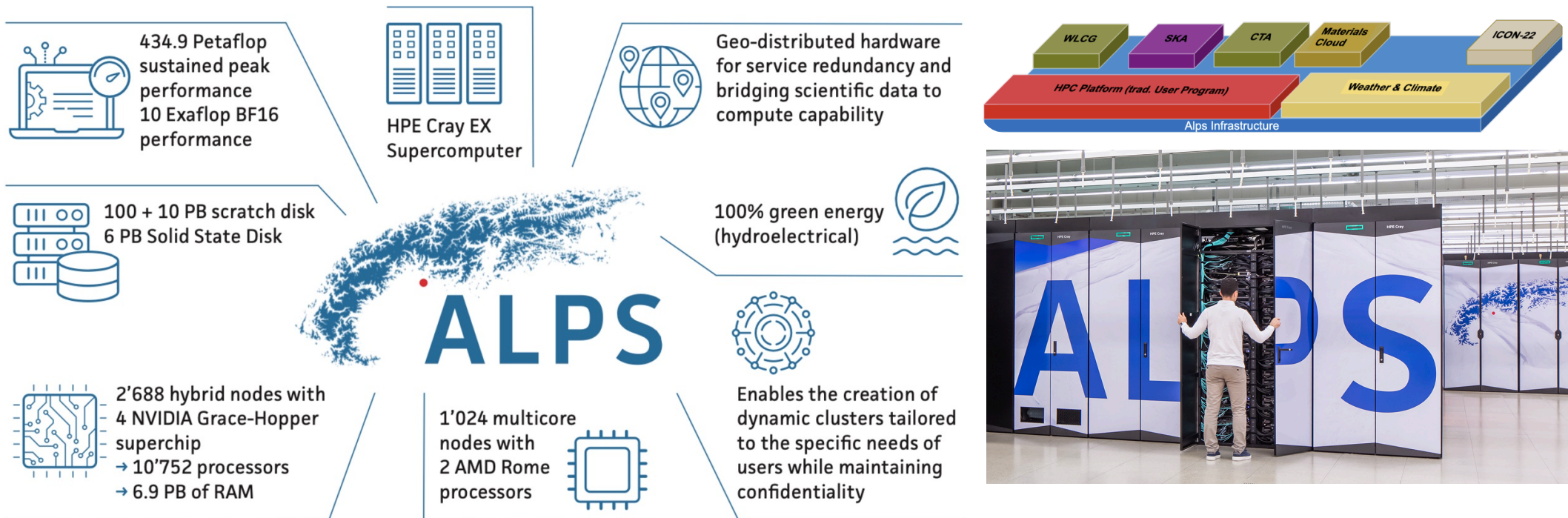


... to Supercomputers: Top Swiss HPC System Alps

Housed at CSCS, EPFL, PSI (data archiving), and ECMWF (Bologna – meteo data)

September 2024 Alps: Cray-HPE Shasta EX **434.9 PFlop/s** (**10^{15} Flop/s**)

HPE Cray Shasta **software stack** (software-defined infrastructure) – “a general-purpose supercomputer”



[<https://www.cscs.ch/computers/alps>]

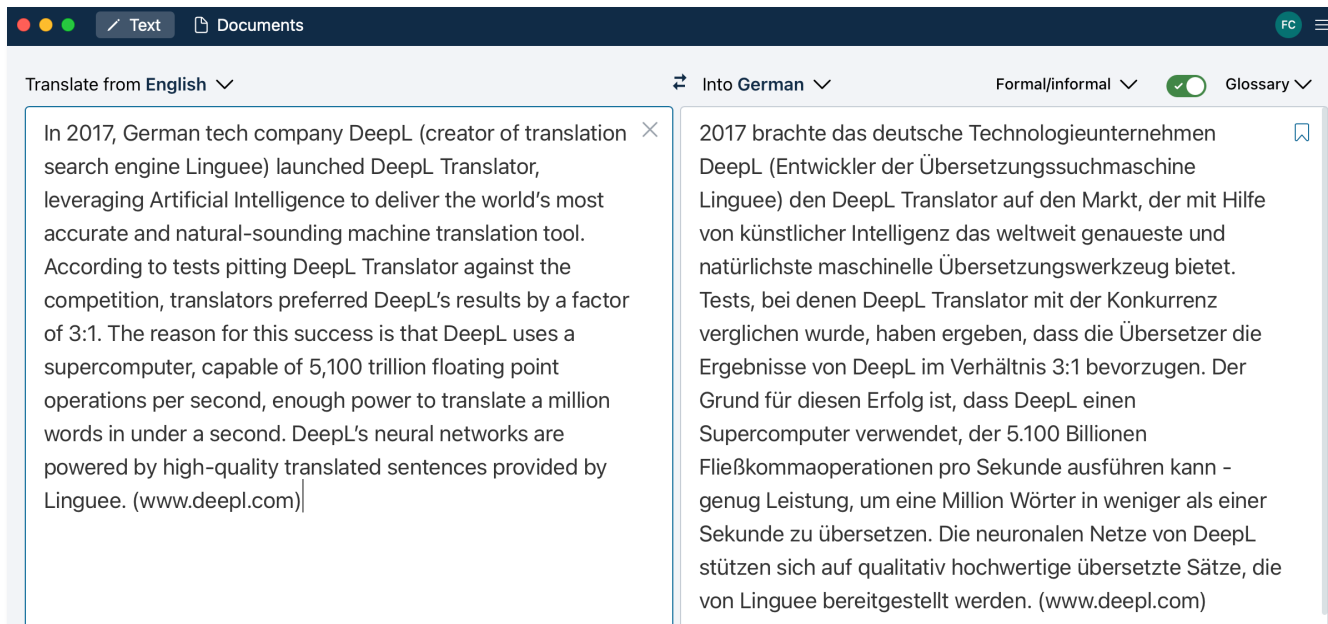
From Everyday Consumer Products ...



- **#HPCMatters to our everyday lives**
- Molecular dynamics simulations for: diapers, shampoo, toothpaste, dyes, perfumes, etc.
- <https://youtu.be/PuCx50FdSic> (2'01")

Everyday Conversations in a Foreign Language ...

AI-powered Translations with DeepL



- **DeepL = Deep Learning + Supercomputing-based translation (5.1×10^{12} Flop/s)**
- Translates 1'000'000 words in less than 1 second on distributed HPC systems

Text from: https://www.eib.org/attachments/pj/financing_the_future_of_supercomputing_en.pdf

Everyday Conversations with an AI ...



what is your name?



Switzerland

⚡ Powered by sovereign compute infrastructure in Lugano,...

Hi, my name is Apertus. I am an AI developed by the Swiss AI Initiative and operated by the Public AI Inference Utility. I serve as an AI assistant to answer questions and provide information across a wide range of topics, from general knowledge to specific domains such as science, coding, and creative writing.

Please feel free to ask me any questions you have. If there's anything I can assist you with today, just let me know!



<https://chat.publicai.co>, <https://www.youtube.com/watch?v=atObT7Xnbdk>,

<https://deepmind.google/science/alphafold/>

Fall 2025

What is the best example of the use of parallel and distributed supercomputing to present to students of a master's course on foundations of distributed systems?



Switzerland

A great example for a master's course on the foundations of distributed systems that illustrates parallel and distributed computing at scale is the [DeepMind/Google's AlphaFold project](#).

Why AlphaFold?

- **Problem Complexity:** Protein folding is a complex computational problem with a high degree of parallelism due to the sheer scale of possible conformations and the long computing time to test all possibilities.
- **Parallel and Distributed Processing:** AlphaFold uses a combination of techniques:
 - **Massive-Scale Parallelization:** Thousands of GPUs were utilized, distributed across multiple clusters, to parallelize the optimization of protein structure predictions. This is critical to handle the problem's computational complexity.
 - **Distributed Trade-offs:** The project effectively distributed the workload across many machines, enabling the leveraging of different hardware configurations. For example, Google's Tensor Processing Units (TPUs) were used in cooperative training to accelerate Deep Learning processes.
- **Real-World Impact:** While traditionally computationally intensive and practically unsolvable for most research groups, the distributed approach successfully predicted the 3D structure of over 200 million proteins in the AlphaFold database.

What It Demonstrates:

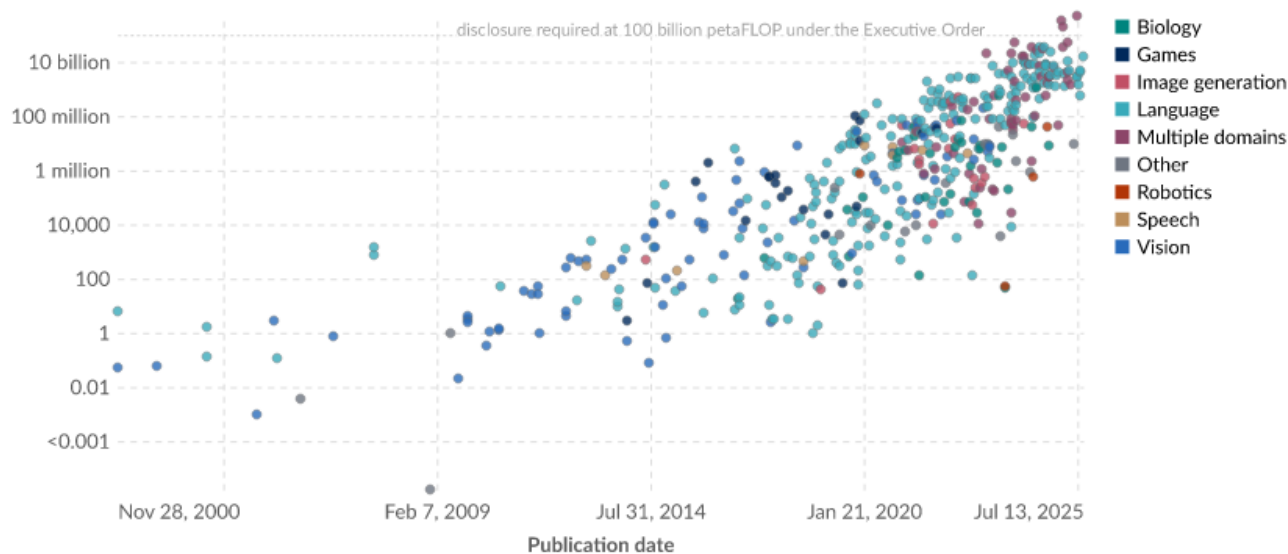
- **Distributed System Design:** Shows how distributing problems across multiple machines can significantly speed up computations.
- **Scalability of Parallel Computing:** Demonstrates the importance of scalability in distributed systems, where the amount of work can increase without significant performance degradation, provided the problem suits parallel execution.
- **Dynamic Workload Management:** Illustrates how different types of machines and computing resources can be utilized effectively for different parts of the computation (e.g., TPUs for neural network training, general-purpose GPUs for matrix operations).
- **Coordinated Systems:** Although AlphaFold's deployment is not fully transparent, discussions around the project often highlight the need for robust distributed job scheduling, data handling, and fault tolerance—key topics in distributed systems studies.

Domain-Specific Conversations with Various AIs ... Trained on Supercomputers

Computation used to train notable artificial intelligence systems, by domain

Computation is measured in total petaFLOP, which is 10^{15} floating-point operations. Estimated from AI literature, albeit with some uncertainty. Estimates are expected to be accurate within a factor of 2, or a factor of 5 for recent undisclosed models like GPT-4.

Training computation (petaFLOP)



Data source: Epoch (2025)

OurWorldinData.org/artificial-intelligence | CC BY

Note: The Executive Order on AI refers to a directive issued by President Biden on October 30, 2023, aimed at establishing guidelines and standards for the responsible development and use of artificial intelligence within the United States.



Training Apertus—Switzerland's first open, multilingual LLM—requires massive computing power. The **#Alps** supercomputer at **#CSCS** enables sovereign AI with GPUs at its core. ETH Zurich's **Ana Klimović** and CSCS's **Joost VandeVondele** explain how. **#apertus #AI #wearealps ETH Zürich EPFL EPFL AI Center**



[https://www.linkedin.com/posts/cscs_alps-cscs-apertus-activity-7373968443525660673-Hcql]

Alps Clariden vCluster

- 1'500 nodes x 4 Nvidia GH200
- Several months (512-1'024 nodes)
- 70B, 15T tokens 6.74×10^{24} Flops
- Production: 3m, 1'024 nodes, 5 GWh

https://github.com/swiss-ai/apertus-tech-report/raw/main/Apertus_Tech_Report.pdf

Everyday Land Transportation ...



https://www.tesla.com/en_eu/modely

A Full Self-Driving (FSD) Computer As a System on a Chip (SoC)

HW4 (& FSD Computer 2)

Samsung Exynos-IP Based

20 CPU Cores

CPU Frequency: 2.35 Ghz

3 Neural Network (NN) Processors

NN Processor Frequency: 2.2 Ghz

Estimated TSMC's 7nm or N4 (4nm class)

50 TOPS

5MP cameras around the car

Tesla Model Y – Hardware 4

16 GB RAM, 256 GB HDD

Everyday Air Transportation ...

Computational Manufacturing and Lifecycle Management

- **Development, testing, manufacturing** of new aircraft involve *extremely high timing and financial risks*
 - *Boeing 787 8y9m from program start to first commercial flight; B777 4y2m*
- Need faster introduction of new aviation technologies, better management associated technological risks
- **Exascale computing** for numerical simulation of aircraft development, production, and operation and their flying impact on atmosphere



[https://www.dlr.de/dlr/en/desktopdefault.aspx/tabid-10204/296_read-23545/#!/gallery/27765]

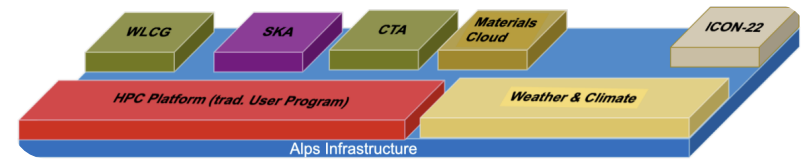
<https://www.hpcwire.com/2019/07/18/aurora-informing-designs-safer-more-efficient-aircraft-exascale-computing/>



July 18, 2019

During the process of designing an aircraft, aeronautical engineers must perform predictive simulations to understand how airflow around the plane

Everyday Weather Forecasting ... In Switzerland!



- **Meteoswiss: ICON** (icosahedral non-hydrostatic model framework) **extreme scale** weather forecast models (thunderstorms, precipitation, temperature, wind, rain, hail) run on an **Alps vCluster**.



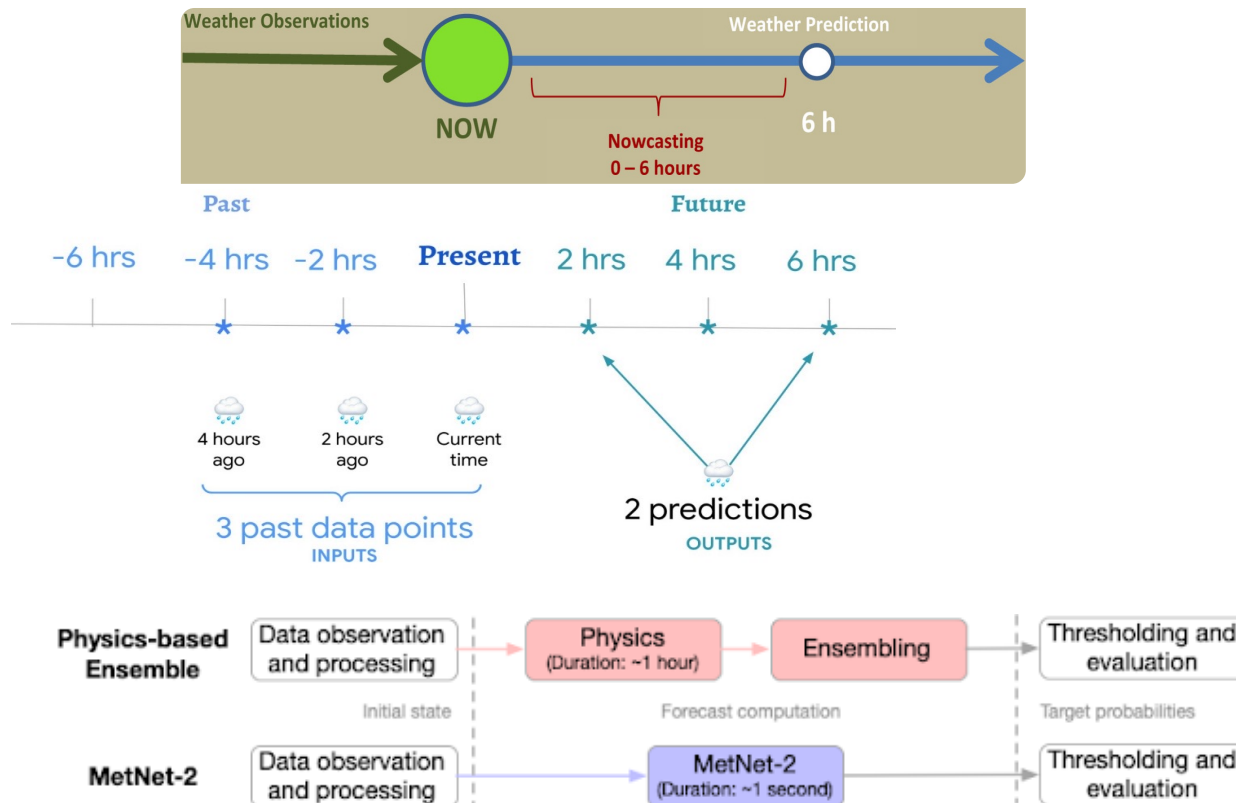
<https://vimeo.com/962909980> (2'40")

<https://www.meteoswiss.admin.ch/weather/warning-and-forecasting-systems/icon-forecasting-systems.html>

- **ICON-CH-EPS** covers entire Alpine region
- **ICON-CH1-EPS** is generated **8 × day** (starting at 00:00) from an ensemble of **11 forecasts**, **1km** grid size, 80 vertical layers (up to 22 km) **up to 33 hours** forecast, at **10-seconds** intervals.
- **ICON-CH2-EPS** is generated **4 × day** (starting at 00:00) from an ensemble of **21 forecasts**, **2.1km** grid size, 80 vertical layers (up to 22 km), **5-day forecast** at **20-seconds** intervals.

Everyday Weather Forecasting ...

Nowcasting using AI/ML



Precipitation nowcasting is high-resolution forecasting of precipitation up to 2-6h ahead.

Outperforms **traditional forecast models** which generate blurry nowcasts at longer lead times.

MetNet-2 a DL model forecasts precipitation at **1 km spatial resolution** and at **2 minutes time resolution**, for up to **12 hours**.

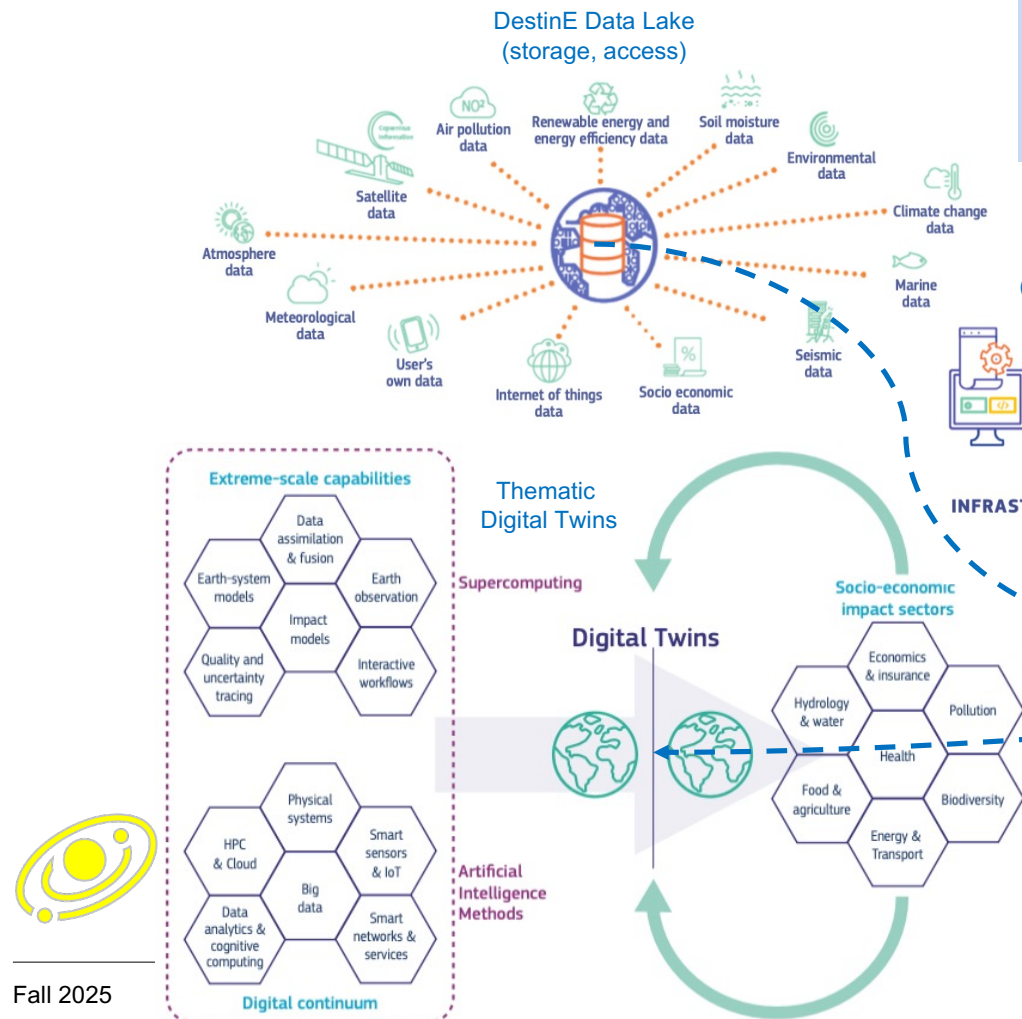
Meteoswiss: ICON + Nowcasting

2023 <https://cloud.google.com/blog/topics/sustainability/weather-prediction-with-ai>.

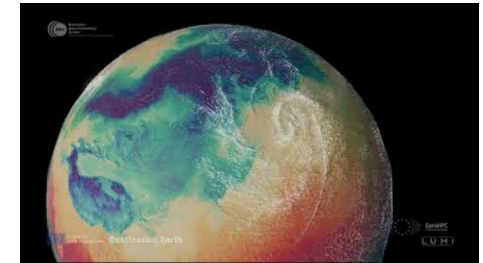
2021 <https://blog.research.google/2021/11/metnet-2-deep-learning-for-12-hour.html>

... to A Whole Digital Twin of Our Earth

Destination Earth



Goal
A 'full' digital replica of the 🌍 by 2030 to address climate change and protect nature.



<https://www.youtube.com/watch?v=hhJW0RBq0hA>



Fall 2025

[Destination Earth Factsheet, 2023,
<https://digital-strategy.ec.europa.eu/en/library/destination-earth-factsheet>]

Videos:
<https://www.youtube.com/watch?v=FKVHZIGqEyw>
<https://youtu.be/vaAfzAMkNGM>
<https://destination-earth.eu/videos-podcasts/>

... and beyond 🌍 all The Way to the ISS for Computing in Space ...

Spaceborne Computer 1 04.2017-04.2019, SBC-2: 02.2021-12.2023, 2.5: 2024- ...

SBC-3: later in this decade on the Moon 🌕



Photo credit: NASA



**Hewlett Packard
Enterprise**

- **Compute-** and **data-**intensive applications (e.g., DNA monitoring) are run on 2 identical Computer systems: SBC on ISS, EBC on 🌍
- **Power consumption** is monitored, and power usage is dynamically tuned during these runs; How to **keep them warm** on Moon w/o air?
- Effects of **radiation** on the systems when running performance applications are determined concurrently with detecting/ analyzing/ adapting to data, quickpath interconnect™ (QPI) internal and FDR external errors
- **SBC-2.5:** 130 TB Kioxia, 1 Nvidia T4GPU, modern OS (RH Linux 7.8), new software, additional security -> use of AI for future lunar missions
- Speeding time-to-insight from months to minutes by minimizing data processing and transfers to/from Earth.

https://www.nasa.gov/mission_pages/station/research/experiments/2304.html#overview, <https://www.youtube.com/watch?v=dJeR6vu3GYE>, https://www.youtube.com/watch?v=tdbie_vceRQ

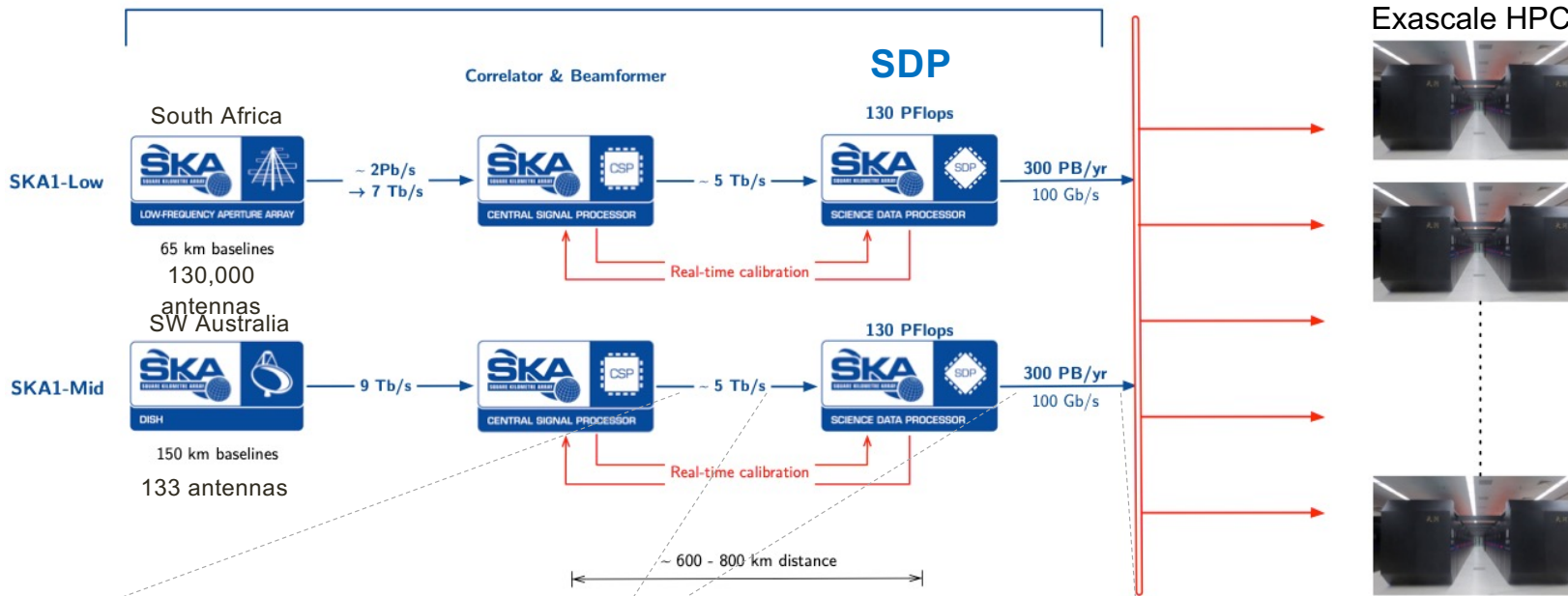
<https://www.extremetech.com/extreme/254089-iss-getting-powerful-computer-ever-sent-space>

<https://www.hpe.com/us/en/newsroom/press-release/2021/02/hewlett-packard-enterprise-accelerates-space-exploration-with-first-ever-in-space-commercial-edge-computing-and-artificial-intelligence-capabilities.html>

<https://spacenews.com/hpes-spaceborne-computer-offers-template-space-based-ai/>

... and Observing and Understating the Evolution of Our Universe

The Square Kilometer Array Observatory



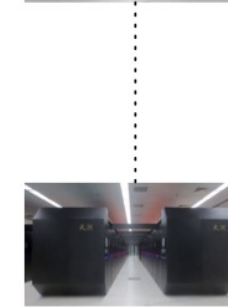
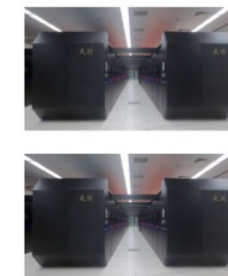
Maintaining **0.5-1 TB/s** data rates out of the correlator to the **SDP** implies **45-85 PB** of raw data / telescope $\Rightarrow$ Cheaper to re-observe data than to store it!

Limited **data delivery rates** from **SDPs** and **data processing rates** by **SRCs** $\Rightarrow$ Reduce the raw data volume at SDP before delivery to users!

SRC

SKA Regional Centres

Exascale HPC



- A Leading **HPC** and **Big Data** Challenge for 2025 and Beyond

[https://indico.in2p3.fr/event/18946/contributions/71400/attachments/53059/68907/SDP_and_SRCs_ESCAPE_WP5.pdf]

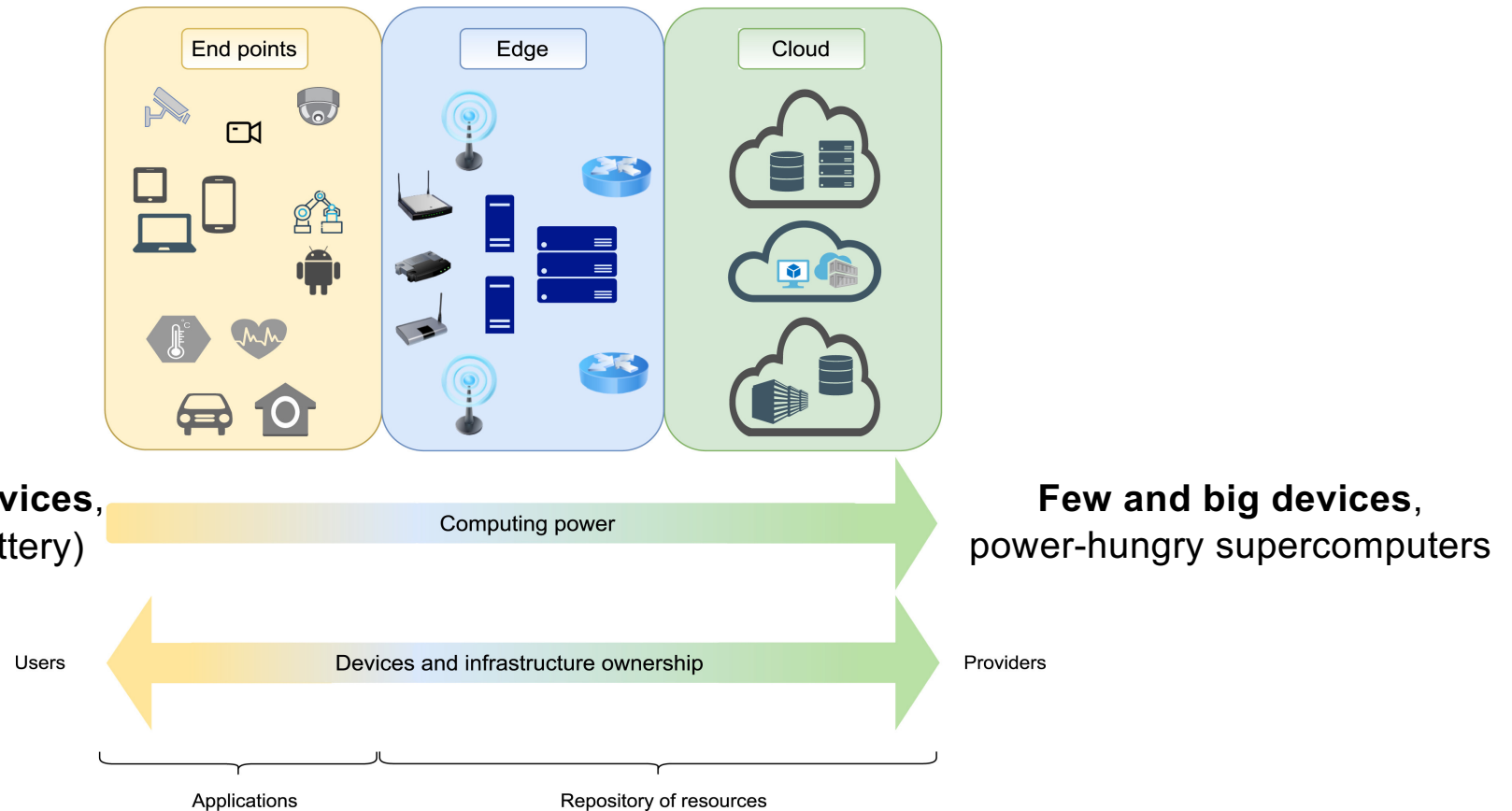
[<https://www.skao.int/en/explore/big-data>]



University of Basel



Applications from a **Processing** Perspective ... Span the Compute Continuum



[Al-Dulaimy, A. *et al.* The computing continuum: From IoT to the cloud. *Internet of Things* **27**, 101272 (2024).]

Applications from a **Processing** Perspective ...

- From a processing perspective, ***we live in an HPC world***, where **supercomputing** is embraced by an increasing spectrum of classical and new fields and markets:

... and many more

